Course Project

Audio Signal Acquisition, Processing, and Quality Evaluation

Course: Signals and Communications systems (CCEN 302) **Total Weight:** 100 Points (100%)

1. Project Overview

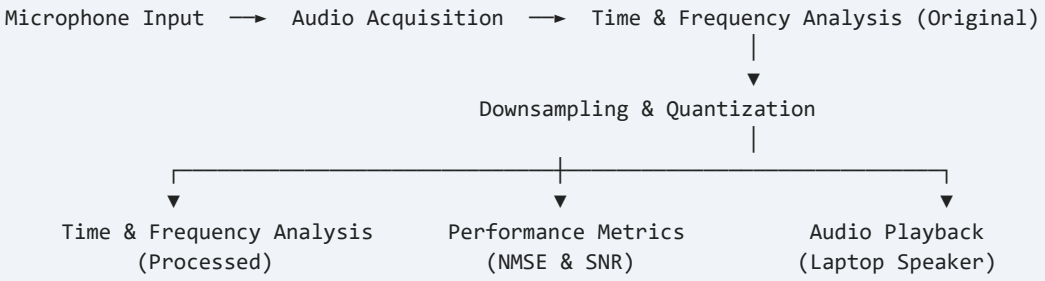
In this hands-on project, students will apply core Signals and Systems concepts: sampling, quantization, SNR/NMSE quality analysis, and time/frequency domain analysis, to real-world acoustic signals. Students will acquire audio via their laptop microphone, process the signal under varying sampling rates and bit depths, measure the resulting quality degradation, analyze the signals in both time and frequency domains, and output the processed audio back through their laptop speakers. Moreover, students will study the quality-storage compromise.

2. Learning Objectives

- **Data Acquisition:** Record real-time continuous audio signals and map them to discrete-time digital representations.
- **Sampling Theory:** Demonstrate the Nyquist-Shannon sampling theorem and observe the visual and auditory effects of aliasing.
- **Quantization Analysis:** Analyze uniform dynamic range quantization and evaluate quantization noise performance.
- **Objective Metrics:** Quantify signal degradation using Normalized Mean Squared Error (NMSE) and Signal-to-Noise Ratio (SNR).
- **Domain Analysis:** Compare original and degraded signals using time-domain waveforms and frequency-domain representations (FFT/Spectrograms).

3. Detailed Project Requirements & Workflow

System Pipeline Overview:



1: Audio Acquisition & Initial Analysis

- **Recording:** Record a 20 seconds speech or music audio clip using your laptop's microphone at a baseline sampling rate fs ≥ 44.1 kHz with 16-bit resolution.
- **Time Domain Analysis:** Plot the continuous-time equivalent waveform $x(t)$ over the recorded duration. You may zoom on certain periods.
- **Frequency Domain Analysis:** Compute and plot the magnitude spectrum $|X(f)|$ using the Fast Fourier Transform (FFT) and generate its spectrogram representation.

2: Sampling Rate Reduction (Downsampling) & Aliasing

- **Processing:** Downsample the recorded signal to multiple lower sampling frequencies (e.g., fs = 22.05 kHz, 11.025 kHz, 8 kHz, and 4 kHz).
- **Aliasing Demonstration:** Compare downsampled signals processed with anti-aliasing low-pass filtering against those downsampled without anti-aliasing filtering. Identify spectral overlap and high-frequency folding in the magnitude spectra.

3: Bit-Depth Reduction (Quantization)

- **Processing:** Re-quantize the original baseline signal from 16-bit resolution down to lower bit depths (e.g., 8-bit, 4-bit, and 2-bit uniform quantization).
- **Quantization Noise Analysis:** Isolate the quantization error signal e(n) = x(n) - x_q(n) and plot its distribution and time-domain behavior.

4: Objective Quality Metrics

Compute objective quality metrics comparing the original baseline signal x(n) to each processed signal x̂(n) across all sampling rate and bit-depth combinations:

1. Normalized Mean Squared Error (NMSE):

$$NMSE = \sum [\, x(n) - \hat{x}(n) \,]^2 \;/\; \sum [\, x(n) \,]^2$$

2. Signal-to-Noise Ratio (SNR in dB):

$$\boldsymbol{SNR\ (dB) = 10 \cdot log_{10}\left(\sum [\, x^2(n) \,] \;/\; \sum [\, x(n) - \hat{x}(n) \,]^2 \right)}$$

5: Audio Output & Perceptual Evaluation

- **Playback:** Export and stream each processed signal x̂(n) to your laptop's speaker.
- **Perceptual Correlation:** Provide a qualitative description matching auditory perception (e.g., harshness, loss of clarity, robotization, background hiss) with objective NMSE/SNR values.
- **Storage:** Evaluate the storage requirements for each sampling and quantization values.

6: Simulink

- Repeat the same experiment using Simulink
- Compare the results obtained using both software.

4. Deliverables

- **Source Code:** Executable script/notebook (MATLAB and Simulink) containing clear comments, modular functions, and reproducible figure generation.
- **Written Technical Report (PDF):** A comprehensive report containing an Executive Summary, Methodological overview, Comparative plots (Time-domain, FFT Magnitude Spectra, Spectrograms), Summary tables of NMSE/SNR across all test conditions, and a Discussion comparing theoretical predictions (e.g., 6 dB/bit rule, Nyquist limit) with experimental results.
- **Audio Files:** A compressed zip folder containing the original recording and re-synthesized output audio files (.wav format).

5. Grading Breakdown (Total: 100%)

Component	Description	Weight
Audio Acquisition & Baseline Analysis	Successful recording, correct time-domain plotting, FFT spectrum generation, and spectrogram setup.	**15%**
Sampling & Aliasing Evaluation	Correct implementation of downsampling (with and without LPF), clear demonstration of spectral aliasing, and frequency plots.	**15%**
Quantization Analysis	Uniform re-quantization across multiple bit-depths (e.g., 16 to 2 bits) and quantization noise characterization.	**15%**
Quantitative Metrics (NMSE & SNR)	Accurate mathematical implementation and tabular comparison of NMSE and SNR across parameters.	**15%**
Audio Output & Perceptual Synthesis	Functional audio playback through laptop speakers and correlation between auditory degradation and metrics.	**10%**
Report Quality & Code Documentation	Professional report formatting, figure labeling, clear code structure, and insightful technical discussion.	**10%**
Q/A		**20%**
Total Grade	**Complete submission satisfying all project requirements.**	**100%**

Guidelines:

1) The project will be implemented in groups of two or four students, no groups of three.
2) The MATLAB part can be implemented with the help of AI
3) Due date: 5/11/2026, at 6:00 PM, firm, no extension.
4) Upload the report in PDF through BB. No submissions by email.
5) MATLAB and Simulink source files should be submitted only through BB and the recorded files will be checked on the laptops directly.

6) All groups mut demonstrate their work in class.